University of Wah

Department of Computer Science

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Course:

Artifical Intelligence_Lab CS-202L

Reg No:

UW-24-CS-BS-101

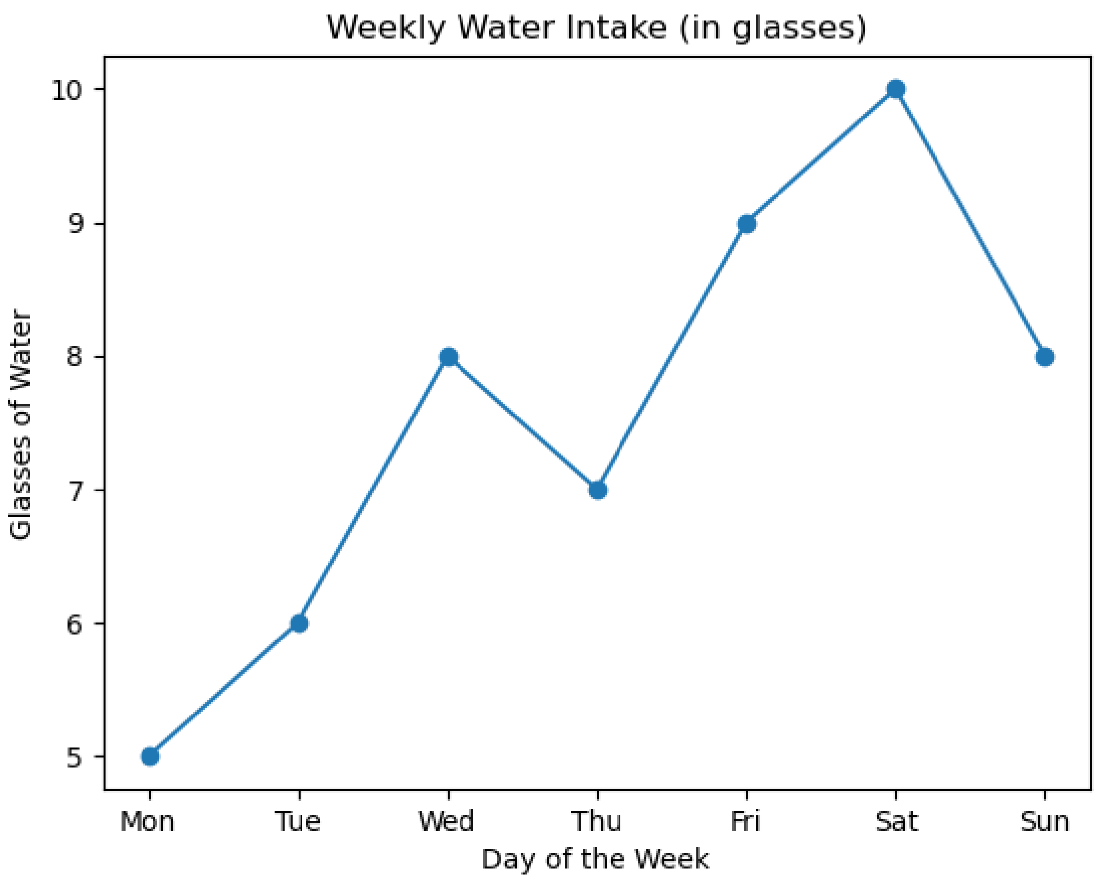
Section:

BS-CS-3rd-B

Task # 01

```
import matplotlib.pyplot as plt
x = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']
y = [5, 6, 8, 7, 9, 10, 8]
plt.plot(x, y, marker='o')
plt.title("Weekly Water Intake (in glasses)")
plt.xlabel("Day of the Week")
plt.ylabel("Glasses of Water")
plt.show()
```

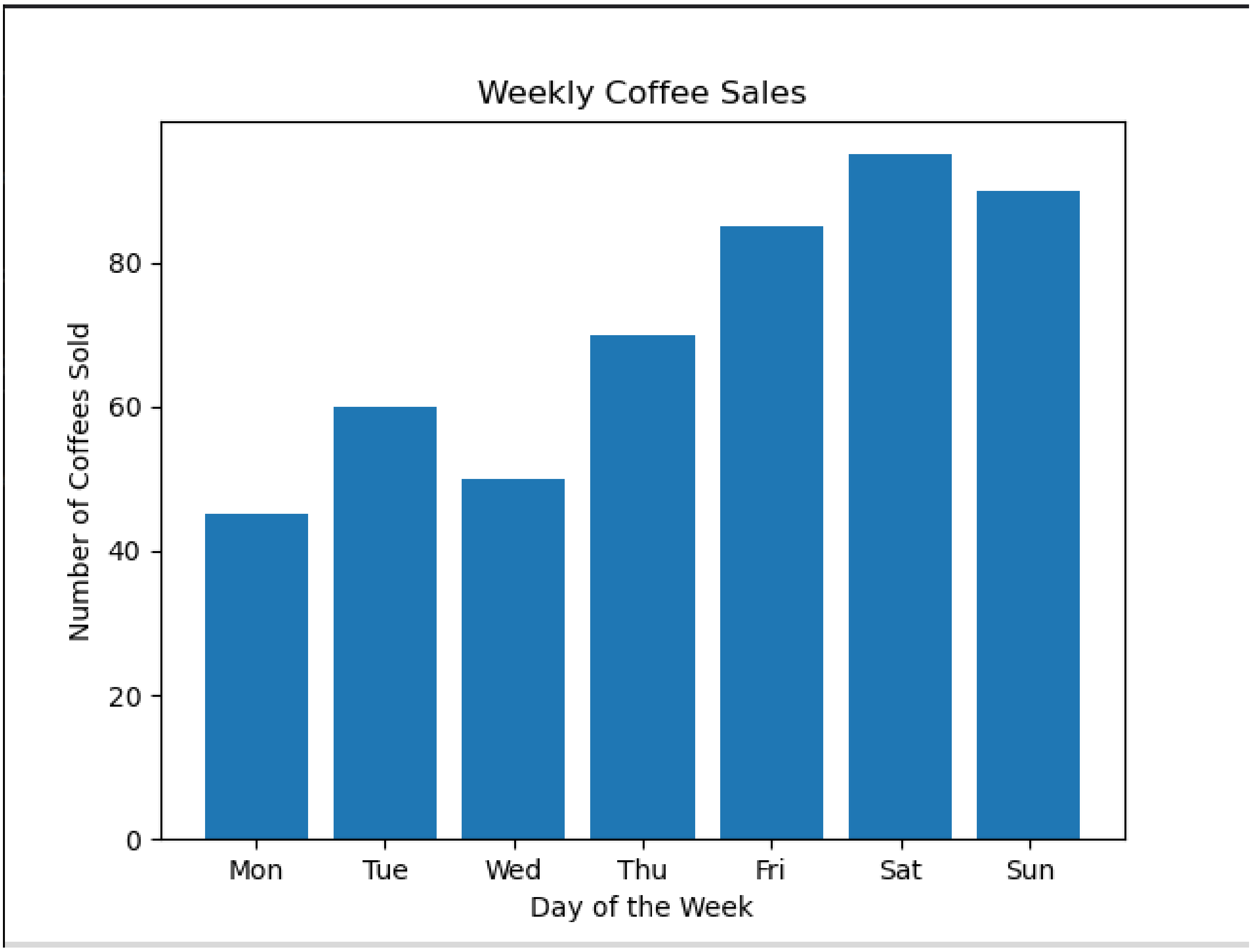
OutPut



Task # 02

```
import matplotlib.pyplot as plt
x = ['Mon', 'Tue', 'Wed', 'Thu', 'Fri', 'Sat', 'Sun']
y = [45, 60, 50, 70, 85, 95, 90]
plt.bar(x, y)
plt.title("Weekly Coffee Sales")
plt.xlabel("Day of the Week")
plt.ylabel("Number of Coffees Sold")
plt.show()
```

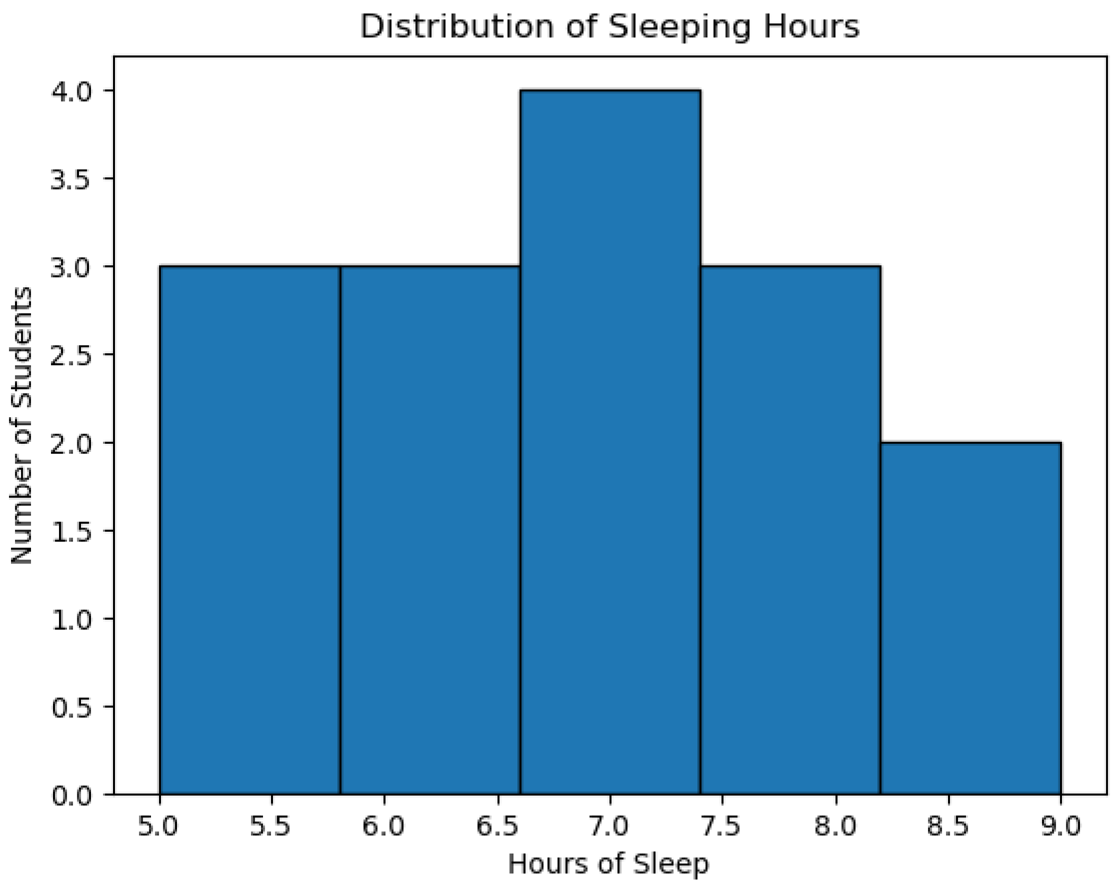
Output



Task # 03

```python
import matplotlib.pyplot as plt
sleep_hours = [5, 6, 7, 8, 5, 6, 7, 7, 8, 9, 5, 6, 8, 9, 7]
plt.hist(sleep_hours, bins=5, edgecolor='black')
plt.title("Distribution of Sleeping Hours")
plt.xlabel("Hours of Sleep")
plt.ylabel("Number of Students")
plt.show()
```

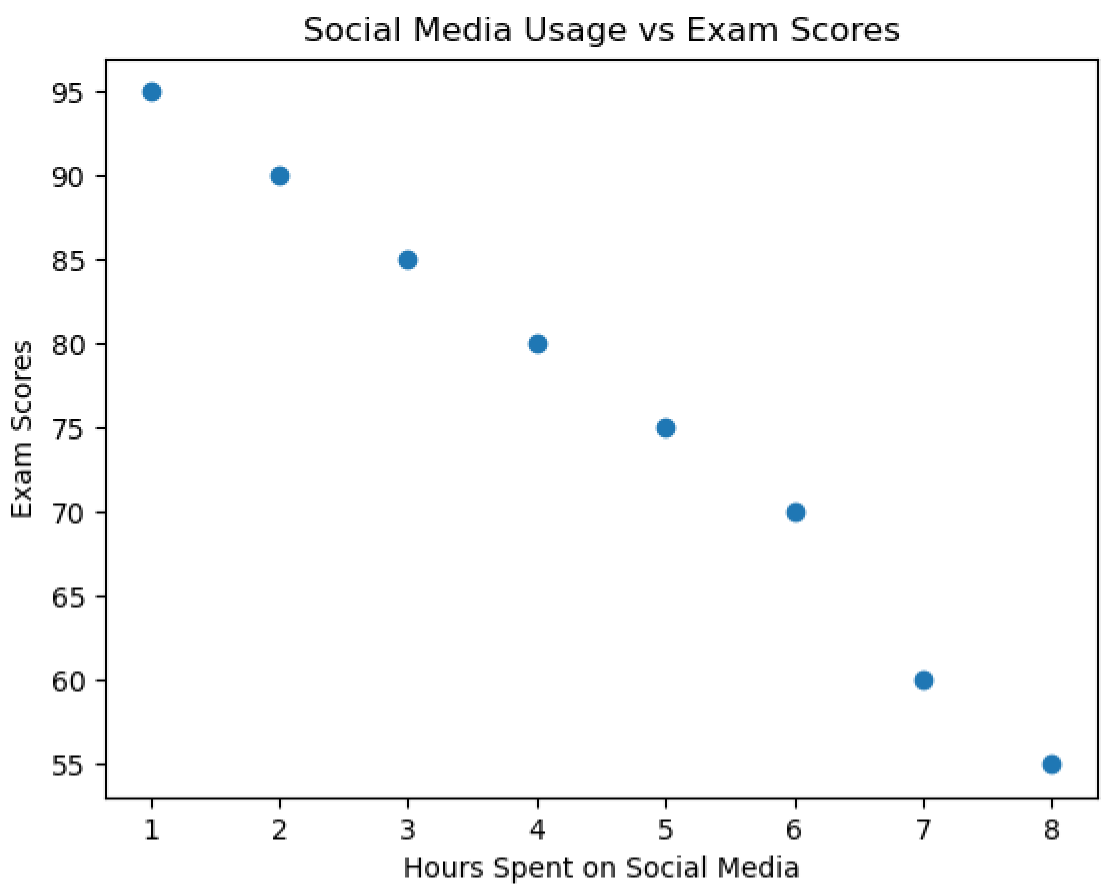
Output



<u>**Task # 04**</u>

```
import matplotlib.pyplot as plt
x = [1, 2, 3, 4, 5, 6, 7, 8]  # Hours on social media
y = [95, 90, 85, 80, 75, 70, 60, 55]  # Exam scores
plt.scatter(x, y)
plt.title("Social Media Usage vs Exam Scores")
plt.xlabel("Hours Spent on Social Media")
plt.ylabel("Exam Scores")
plt.show()
```

<u>**Output**</u>



Task # 05

```python
import matplotlib.pyplot as plt
categories = ['Food', 'Rent', 'Transport', 'Shopping', 'Savings']
expenses = [25, 35, 15, 10, 15]
plt.pie(expenses, labels=categories, autopct='%1.1f%%')
plt.title("Monthly Expenses Breakdown")
plt.show()
```

Output